

cp_4a

1.

Find the Remainder | 🟡 20/20

Easy

Write a program that reads two numbers `A` and `B`, and prints the remainder when `A` is divided by `B`.

Input

The first line of input contains an integer representing `A`.
The second line of input contains an integer representing `B`.

Output

The output should be a single line containing an integer that is the remainder when `A` is divided by `B`.

Explanation

For example, if the given numbers are `A = 10` and `B = 3`,

- The remainder is `1` when `10` is divided by `3`.

The output should be `1`.

Sample Input 1

```
10
3
```

Sample Output 1

```
1
```

Sample Input 2

```
24
5
```

Sample Output 2

```
4
```

code:

```
a=int(input())
b=int(input())
rem=a%b
print(rem)
```

2.

Divisible by Seven | 🟡 20/20

● Solved

Easy

Write a program that reads a number `N` and checks if the number `N` is divisible by 7.

Print **Divisible by Seven** if `N` is divisible by 7. Otherwise, print **Not Divisible by Seven**.

Note

A Number `N` is divisible by 7, if the remainder is 0 when `N` is divided by 7.

Example:

- 35 is divisible by 7. (Remainder is 0 when 35 is divided by 7)
- 27 is not divisible by 7. (Remainder is 6 (not 0) when 27 is divided by 7)

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing a string. **Divisible by Seven** should be printed if the number is divisible by 7. Otherwise, **Not Divisible by Seven** should be printed.

Explanation

For example, if the given number `N = 35`,

- The remainder is 0 when 35 is divided by 7.

The output should be **Divisible by Seven**.

Sample Input 1

35

Sample Output 1

Divisible by Seven

Sample Input 2

8

Sample Output 2

Not Divisible by Seven

code:

```
n=int(input())
rem=n%7
if(rem==0):
    print("Divisible by Seven")
else:
    print("Not Divisible by Seven")
```

3.

Quotient & Remainder | 🟡 20/20

Easy

Write a program that reads two numbers `A` and `B` and prints the Quotient and Remainder when `A` is divided by `B`.

Note

For Example, if `5` is divided by `2` (`5/2`),

`Divisor ← 2` `5` `2` `→ Quotient`

`4`

`1` `→ Remainder`

- Quotient is `2` (Quotient should be an integer).
- Remainder is `1`

Input

The first line of input contains an integer representing `A`.

The second line of input contains an integer representing `B`.

Output

The first line of output should be an integer that is the Quotient.

The second line of output should be an integer that is the Remainder.

Explanation

For example, if the given numbers are `A = 5` and `B = 2`,

- The Quotient is `2` when `5` is divided by `2`.
- The Remainder is `1` when `5` is divided by `2`.

The output should be,

```
2
1
```

Sample Input 1

```
5
2
```

Sample Output 1

```
2
1
```

code:

```
a=int(input())
b=int(input())
quotient=a/b
```

```
remainder=a%b
print(int(quotient))
print(remainder)
```

4.

Even or Odd | 🏆 20/20

Write a program that reads a number `N` and checks if `N` is divisible by `2`.

Print **Even** if `N` is divisible by `2`. Otherwise, print **Odd**.

Note

A Number `N` is divisible by `2`, if the remainder is `0` when `N` is divided by `2`.

Example:

- `4` is divisible by `2`. (Remainder is `0` when `4` is divided by `2`)
- `3` is **not** divisible by `2`. (Remainder is `1` (**not 0**) when `3` is divided by `2`)

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing a string. **Even** should be printed if `N` is divisible by `2`. Otherwise, **Odd** should be printed.

Explanation

For example, if the given number is `N = 4`,

- The remainder is `0` when `4` is divided by `2`.

The output should be **Even**.

For example, if the given number is `N = 3`,

- The remainder is `1` (**not 0**) when `3` is divided by `2`.

The output should be **Odd**.

Sample Input 1

4

Sample Output 1

Even

Sample Input 2

3

Sample Output 2

Odd

code:

```
number=int(input())
if(number%2==0):
    print("Even")
else:
    print("Odd")
```

5.

Greatest Among Remainders | 25/25

Solved

Easy

Write a program that reads a number `N` and finds the,

- Remainder when `N` is divided by 4.
- Remainder when `N` is divided by 5.

Print the greatest remainder among the two remainders when `N` is divided by 4 and 5.

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing an integer that is the greatest remainder among the two remainders when `N` is divided by 4 and 5.

Explanation

For example, if the given number is `N = 12`,

- The remainder is 0 when `N` is divided by 4.
- The remainder is 2 when `N` is divided by 5.
- The greatest remainder among the two remainders 0 and 2 is 2.

The output should be 2.

Sample Input 1

12

Sample Output 1

2

Sample Input 2

147

Sample Output 2

3

code:

```
n=int(input())
rem_4=n%4
```

```
rem_5=n%5
if(rem_5>rem_4):
    print(rem_5)
else:
    print(rem_4)
```

6.

Special Eleven 🟡 25/25

● Solved

Write a program that reads a number `N` and checks if the remainder is `0` or `1` when `N` is divided by `11`.

Print **Special Eleven** if the remainder is `0` or `1` when `N` is divided by `11`. Otherwise, print **Normal Number**.

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing a string. **Special Eleven** should be printed if the remainder is `0` or `1` when `N` is divided by `11`. Otherwise, **Normal Number** should be printed.

Explanation

For example, if the given number is `N = 22`,

The output should be **Special Eleven** as the remainder is `0` when `N` is divided by `11`.

For example, if the given number is `N = 23`,

The output should be **Special Eleven** as the remainder is `1` when `N` is divided by `11`.

For example, if the given number is `N = 15`,

- The remainder is `4` when `N` is divided by `11`. (`4` is **not** equal to `0` or `1`)

The output should be **Normal Number** as the remainder is `4` (**not** equal to `0` or `1`) when `N` is divided by `11`.

Sample Input 1

22

Sample Output 1

Special Eleven

Sample Input 2

23

Sample Output 2

Special Eleven

Sample Input 3

15

Sample Output 3

Normal Number

code:

```

n=int(input())
rem_1=n%11==1
rem_0=n%11==0
if(rem_0 or rem_1):
    print("Special Eleven")
else:
    print("Normal Number")

```

7.

2 Digit Divisible Number 🟡 25/25

● Solved

Easy

Write a program that reads a two-digit number N and checks if N is divisible by both the digits of N .
Print the double of N if N is divisible by both the digits of N . Otherwise, print N .

Note

The double of N is calculated as $N * 2$.

Example: $N = 15$

- The double of 15 is 30 ($15 * 2 = 30$).

Input

The input will be a single line containing an integer representing N .

Output

The output should be a single line containing an integer. The double of N should be printed if N is divisible by both the digits of N . Otherwise, N should be printed.

Explanation

For example, if the given number is $N = 15$,

The digits of N are 1 and 5.

✓ N is divisible by 1. (15 is divisible by 1)

✓ N is divisible by 5. (15 is divisible by 5)

As N is divisible by both the digits of N , print the double of N .

The output should be 30 as the double of N is 30 ($15 * 2 = 30$)

For example, if the given number is $N = 26$,

The digits of N are 2 and 6.

✓ N is divisible by 2. (26 is divisible by 2)

✗ N is divisible by 6. (26 is **not** divisible by 6)

As N is **not** divisible by both the digits of N , print N .

The output should be 26.

Sample Input 1

15

Sample Output 1

30

code:

```

n=input()
num_1=int(n)%int(n[0])==0
num_2=int(n)%int(n[1])==0
if(num_1 and num_2):
    print(int(n)*2)
else:
    print(n)

```

8.

Lucky Number - 2 🟡 30/30

● Solve

Write a program that reads a two-digit number `N` and checks if any of the given conditions is satisfied.

- The number `N` is divisible by **9**.
- One of the digits of the number `N` is equal to **9**.

Print **Lucky Number** if any of the given conditions is satisfied. Otherwise, print **Unlucky Number**.

Note

The number `N` is divisible by **9**, if the remainder is **0** when `N` is divided by **9**.

Input

The input will be a single line containing an integer representing `N`.

Output

The output should be a single line containing a string. **Lucky Number** should be printed if any of the given conditions is satisfied. Otherwise, **Unlucky Number** should be printed.

Explanation

For example, if the given number is `N = 18`,

- ✓ The number `N` is divisible by **9**. (**18** is divisible by **9**)
- ✗ One of the digits of the number `N` is equal to **9**. (**1, 8** are **not** equal to **9**)

The output should be **Lucky Number** as one of the given conditions is satisfied.

Sample Input 1

18

Sample Output 1

Lucky Number

Sample Input 2

13

Sample Output 2

Unlucky Number

code:

```
num=input()  
num_0=int(num[0])==9  
num_1=int(num[1])==9  
n=int(num)  
if(n%9==0 or (num_1 or num_0)):  
    print("Lucky Number")  
else:  
    print("Unlucky Number")
```

9.

Special String | 30/30

Solved

Easy

Write a program that reads a string `S` and checks if all the given conditions are satisfied.

- The first three characters of `S` is **NXT**.
- The remaining characters of `S` contain a Number. Number is divisible by 2 or 7.

Print **Special String** if all the given conditions are satisfied. Otherwise, print **Not a Special String**.

Quick Tip

Make sure the number is converted to an **integer** before doing any operations like Division (`/`), Remainder (`%`), etc.

Example: `S = NXT1234`

- The number in `NXT1234` is 1234.
- 1234 should be converted to an integer before doing any operations.

Input

The input will be a single line containing a string representing `S`.

Output

The output should be a single line containing a string. **Special String** should be printed if all the given conditions are satisfied. Otherwise, **Not a Special String** should be printed.

Explanation

For example, if the given string is `A = NXT1234`,

- ✓ The first three characters of `S` is **NXT**. (The first three characters of `NXT1234` is equal to **NXT**)
- ✓ The remaining characters of `S` is divisible by 2 or 7. (The remaining characters of `NXT1234` is 1234. 1234 is divisible by 2)

The output should be **Special String** as all the given conditions are satisfied.

Sample Input 1

```
NXT1234
```

Sample Output 1

```
Special String
```

Sample Input 2

```
PRA49
```

Sample Output 2

```
Not a Special String
```

code:

```
string=input()
first_half=string[:3]
second_half=int(string[3:])
cond_1=first_half=="NXT"
cond_2=second_half %2==0 or second_half%7==0
if(cond_2 and cond_1):
    print("Special String")
```

```
else:
    print("Not a Special String")
```

10.

Pair of Numbers - 2 | 🟡 30/30

● Solved

Easy

Write a program that reads two numbers `A` and `B`, and checks if all the given conditions are satisfied.

- `A` and `B` are divisible by 3.
- `A` or `B` is divisible by 12.

Print **Pair** if all the given conditions are satisfied. Otherwise, print **Not a Pair**.

Input

The first line of input contains an integer representing `A`.
The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a string. **Pair** should be printed if all the given conditions are satisfied. Otherwise, **Not a Pair** should be printed.

Explanation

For example, if the given numbers are `A = 15` and `B = 240`,

- ✓ `A` and `B` are divisible by 3. (15 and 240 are divisible by 3)
- ✓ `A` or `B` is divisible by 12. (240 is divisible by 12)

The output should be **Pair** as all the given conditions are satisfied.

Sample Input 1

```
15
240
```

Sample Output 1

```
Pair
```

Sample Input 2

```
360
7
```

Sample Output 2

```
Not a Pair
```

code:

```
a=int(input())
b=int(input())
cond1=a%3==0 and b%3==0
cond2=a%12==0 or b%12==0
if(cond2 and cond1):
    print("Pair")
```

```
else:
    print("Not a Pair")
```

11.

Lucky Number - 3 30/30

 Solved

Write a program to print if the given number is divisible by any of the lucky numbers 6, 3, 2 in decreasing order of priority(6 is luckier than 3 and 3 is luckier than 2).
Print "Number is divisible by" followed by the luckiest number among the above 3 which can divide the number.
Print "Number is not divisible by 2, 3 or 6" if the number is not divisible by any of them.

Input

The input will be a single line containing a positive number.

Output

If the given number is divisible by 6, print "Number is divisible by 6".
If the given number is divisible by 3, print "Number is divisible by 3".
If the given number is divisible by 2, print "Number is divisible by 2".
In all other cases print "Number is not divisible by 2, 3 or 6".

Explanation

In the example 126 is divisible by 2, 3 and 6
But 6 takes precedence because 6 is luckiest amongst the three. So the output should be "Number is divisible by 6"

Sample Input 1

126

Sample Output 1

Number is divisible by 6

Sample Input 2

27

Sample Output 2

Number is divisible by 3

Sample Input 3

55

Sample Output 3

Number is not divisible by 2, 3 or 6

code:

```
num=int(input())
if(num%6==0):
    print("Number is divisible by 6")
if(num%3==0) and num%6!=0:
    print("Number is divisible by 3")
if(num%2==0) and num%6!=0:
    print("Number is divisible by 2")
```

```
if(num%2!=0 and num%3!=0 and num%6!=0):
    print("Number is not divisible by 2, 3 or 6")
```

12.

Special Number 👤 30/30

● Solved

Write a program that reads a two-digit number `N` and checks if any of the given conditions is satisfied.

- The sum of digits of `N` is equal to 7.
- One of the digits of `N` is equal to 7.
- `N` is divisible by 7.

Print **Special Number** if any of the given conditions is satisfied. Otherwise, print **Normal Number**.

Input

The input will be a single line containing a two-digit integer representing `N`.

Output

The output should be a single line containing a string. **Special Number** should be printed if any of the given conditions is satisfied. Otherwise, **Normal Number** should be printed.

Explanation

For example, if the given two-digit number is `N = 67`

- ✗ The sum of digits of `N` is equal to 7. ($6 + 7 = 13$, 13 is **not** equal to 7)
- ✓ One of the digits of `N` is equal to 7. (6, 7 are digits of 67. 7 is equal to 7)
- ✗ `N` is divisible by 7. (67 is **not** divisible by 7)

The output should be **Special Number** as one of the given conditions is satisfied.

Sample Input 1

```
67
```

Sample Output 1

```
Special Number
```

Sample Input 2

```
36
```

Sample Output 2

```
Normal Number
```

code:

```
n=input()
cond1=int(n[0])+int(n[1])==7
cond2=int(n[0])==7 or int(n[1])==7
cond3=int(n)%7==0
if(cond1 or cond2 or cond3):
    print("Special Number")
```

```
else:  
    print("Normal Number")
```